

Machine Learning–Guided Screening of Magnetic Materials for Thermally Robust Non-Volatile Data Storage

Full project report — methodology, figures, and results across all five phases

Prepared for advisor review · 21 August 2026

How to read this report

Each phase section states what was done, exactly how (data source, method, tool), and what the result was, with the supporting figure or table inline. Section 6 (DFT validation) is the newest work and the main discussion item for this meeting.

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1. Project goal and pipeline overview

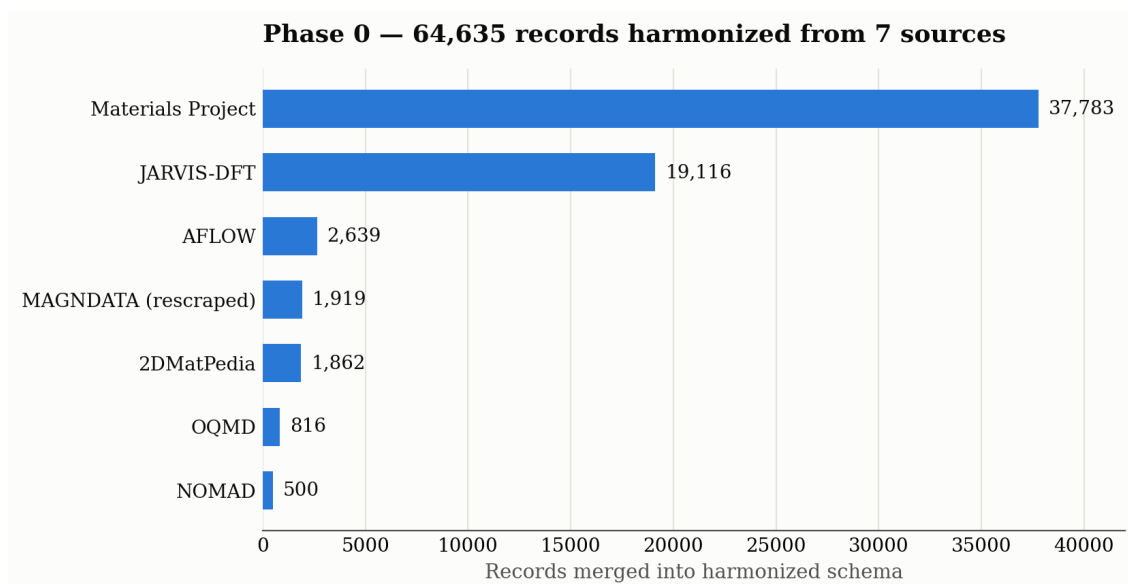
The project screens magnetic materials — computationally and, where candidates warrant it, via spin-dynamics simulation — for systems that host stable, controllable spin textures (e.g. spin spirals, skyrmion-hosting interfaces such as Pd/Fe/Ir(111)-type stacks) suitable for thermally robust non-volatile memory. The approach combines (a) a large harmonized materials database, (b) an ML classifier for magnetic ordering that corrects for known sampling bias in existing databases, (c) a machine-learned interatomic potential (MLIP) fine-tuned for magnetic systems, and (d) atomistic spin-dynamics simulation of shortlisted candidates, cross-checked against real DFT.

Phase	Purpose	Status
0 — Data harmonization	Merge multiple public materials databases into one schema	Complete — 7 sources, 64,635 records
1 — Ordering classifier	Predict magnetic ordering from composition + structure	Complete — v2, 67.0% accuracy
2 — Candidate screening	Filter for DMI-compatible candidates, backfill structures, relax with MACE	Complete — 298 relaxed, top-30 shortlist
3 — MLIP fine-tuning	Fine-tune MACE for fast, accurate magnetic-system energetics	Complete — 60/60 epochs
4 — Spin dynamics & DFT validation	LAMMPS-SPIN simulation + real VASP energy mapping	In progress — 1 system simulated, 24 of 26 candidates DFT-validated

2. Phase 0 — Data harmonization

PHASE 0 · COMPLETE

All sources are merged into one harmonized schema (`material_id`, `formula`, `elements`, `space_group`, `density`, `volume`, `band_gap`, `ordering`, `total_magnetization`, and — where available — full structure and per-site `magmom` vectors). Records keep their source database tagged rather than being silently deduplicated.



Source	Records	Fields	Notes
Materials Project	37,783	full schema	via <code>mp-api</code>
JARVIS-DFT	19,116	full schema	0% populated ordering labels (see §3)
AFLOW	2,639	full schema	
MAGNDATA (rescaped)	1,919 of 2,414 attempted	full schema + structure + per-site <code>magmom</code>	only ground-truth magnetic-structure source
2DMatPedia	1,862	full schema + structure	project's only 2D-materials source so far
OQMD	816	composition + stability	no magnetism fields; cross-validation role only
NOMAD	500	full schema	
Total	64,635	3,781 with full structure	

2.1 MAGNDATA rescrape

MAGNDATA (Bilbao Crystallographic Server) is the only source with **ground-truth** magnetic structure (not a composition-based guess) — a hard blocker for both the Phase 1 classifier and Phases 3–4, which need per-site magnetic moments as input. The existing scrape covered only 79–102 records with no surviving acquisition script. A new scraper was written and verified end-to-end against the live server: the full index returns all 2,414 unique entries in one page (no pagination, $\sim 24\times$ the previous scrape); structure files (MCIF) are downloaded once the filename-construction rule was reverse-engineered; per-site magnetic moments, space group, and structure are parsed via `pymatgen`, with ordering (FM/AFM/FiM/NM) derived from net-vs-summed moment magnitude and validated against a known case (LaMnO_3 , correctly recovered as AFM, space group 62/Pnma).

Result

1,919 / 2,414 entries successfully parsed (79.5%). Of the 495 permanent failures, 466 are disordered/partial-occupancy structures that `pymatgen`'s CIF parser cannot represent — an upstream limitation, not a scraper bug.

2.2 Data-quality bug found and fixed

OQMD's `elements` and `space_group_number` fields were silently hardcoded to `None` in the acquisition script despite the raw source fields being present and parseable — discovered because the Phase 2 screening funnel was dropping exactly 816 records, OQMD's full count. Fixed via `pymatgen.core.Composition` and `pymatgen.symmetry.groups.SpaceGroup` (with an underscore-notation fallback for screw-axis symbols like "P21/m" $\rightarrow$ "P2_1/m"); this unblocked 32 OQMD candidates in the Phase 2 funnel.

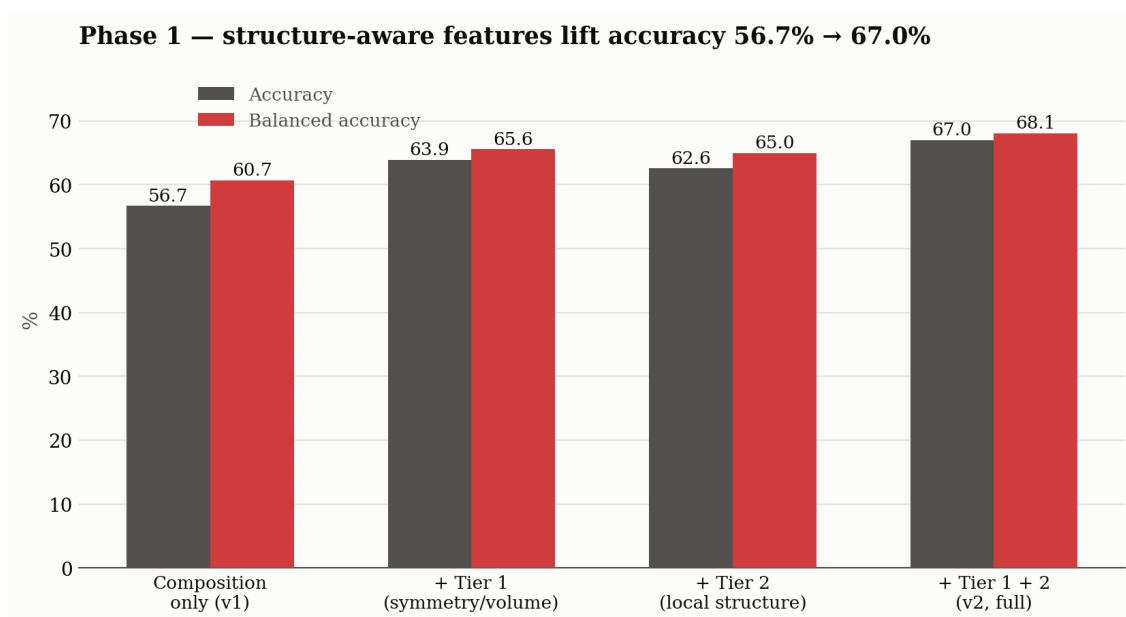
3. Phase 1 — Bias-corrected magnetic-ordering classifier

PHASE 1 · COMPLETE

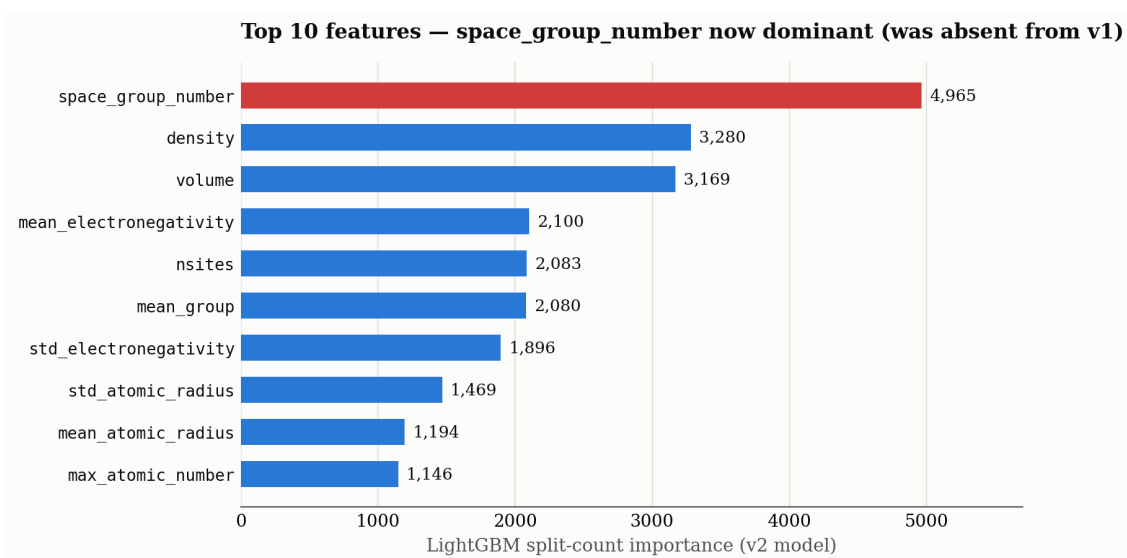
A LightGBM model predicts ordering (FM/AFM/FiM/NM) for any candidate. The v1 baseline used only composition-derived features (per-element statistics: mean/min/max/range/std of atomic number, mass, electronegativity, atomic radius, plus group/period/transition-metal/rare-earth fractions). Retraining on 24× more MAGNDATA ground-truth data left v1's accuracy essentially flat (56.7%) — a real ceiling, because the feature set never consumed structure or per-site `magmom` fields that the new records actually carried.

3.1 Structure-aware features (v2)

Two tiers of structural features were added. **Tier 1** (broad, ~100% coverage): space group number, crystal-system bucket, unit-cell volume, site count. **Tier 2** (narrow, MAGNDATA-only, ~5.3% coverage): local geometry around magnetic-species sites (nearest magnetic-magnetic neighbor distance, coordination number) — computed from atomic positions and species only, deliberately *not* from the `magmom` vectors themselves, since that would leak the label-generating function. Missing Tier 2 values are left as `NaN`; LightGBM handles this natively.



Both tiers contribute independently: broad symmetry/volume fields close roughly two-thirds of the gap on their own; the narrow, MAGNDATA-only local-geometry features add a comparable amount despite covering only ~5% of rows. `space_group_number` is now the single most important feature (previously absent from v1 entirely).



Metric	v1 (composition only)	v2 (full, current)
Accuracy	56.7%	67.0%
Balanced accuracy	60.7%	68.1%
Train / test size	27,234 / 6,809	28,769 / 7,193
Class balance	FM 22,259 · FiM 6,238 · AFM 3,735 · NM 3,730	

Material finding, relevant to all downstream phases

JARVIS (19,116 raw records, would be the second-largest source) had **zero** populated `ordering` labels. Of 61,457 harmonized records, only 35,962 had a usable ordering label at all — Materials Project (34,037) and MAGNDATA (1,919) account for effectively all of it. The Phase 2 deployment classifier now predicts ordering for these previously-unusable records, but their labels are model *predictions*, not ground truth — distinct from the `ground_truth`-sourced labels this report's Phase 4b DFT check focuses on.

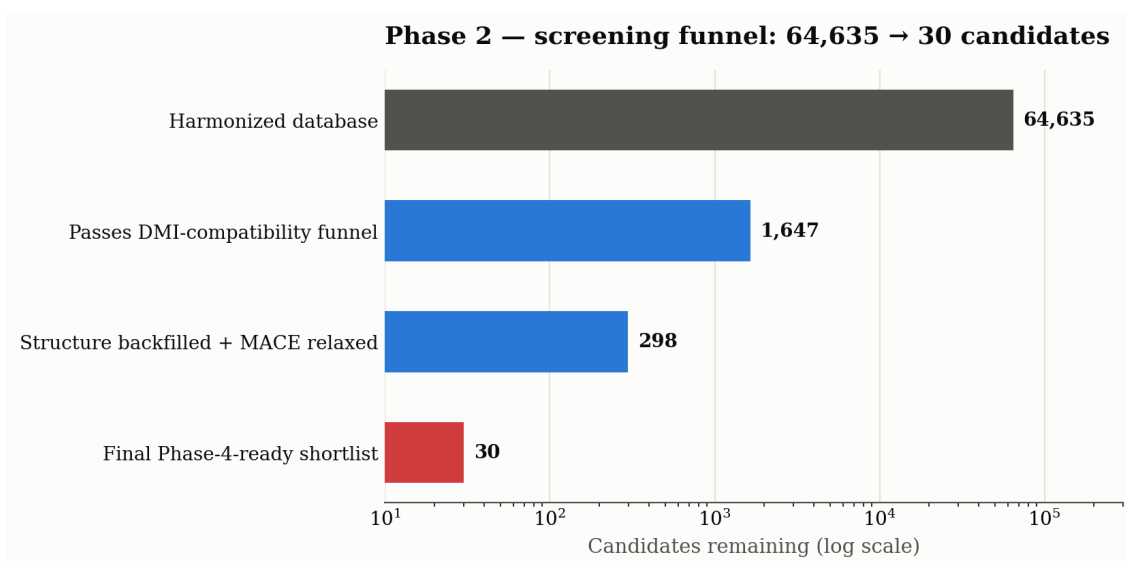
4. Phase 2 — Candidate screening, structure backfill, MACE relaxation

PHASE 2 · COMPLETE

A deployment classifier (same v2 features, trained on all labeled data) predicts ordering for every record lacking a ground-truth label, including all 19,116 JARVIS records. The full 64,635-record database is then filtered for candidates physically compatible with a Dzyaloshinskii–Moriya interaction (DMI) — the mechanism responsible for stabilizing the chiral spin textures this project targets.

Screening criteria

- **Net magnetic moment** — ordering is FM or FiM (AFM excluded; the Phase 4 simulation code assumes a net-moment texture).
- **Noncentrosymmetric space group** — DMI is symmetry-forbidden under inversion. Verified computationally: all 230 space groups checked for an inversion operator (92 centrosymmetric / 138 noncentrosymmetric, matching the textbook count).
- **A heavy spin–orbit-coupling element present** (Ta, W, Re, Os, Ir, Pt, Au, Hg, Bi) — the classic DMI-enhancing species (Fert–Levy / Moriya mechanisms). A heuristic compositional filter, not a computed DMI magnitude.
- **Thermodynamic plausibility** — `energy_above_hull` ≤ 0.1 eV/atom where known (92.6% coverage).



Of 64,635 records, **1,647** pass the full funnel (MP 772, AFLOW 386, JARVIS 353, NOMAD 62, 2DMatPedia 40, OQMD 32, MAGNDATA 2).

4.1 Structure backfill and MACE relaxation

For the top 100 screened candidates per source, structures were fetched via each source's own public data-access method (JARVIS's bulk figshare dataset, AFLOW's AFLUX REST API, Materials Project's `mp_api`) — 299 of 300 attempted backfills succeeded. Each backfilled structure, already DFT-relaxed at its source's own settings, was then re-relaxed with the fine-tuned MACE potential (position-only / fixed-cell) as an independent consistency check.

Result

298/298 candidates relaxed without error and converged (max force ≤ 0.05 eV/Å). Relaxation-energy shift was near-zero for the overwhelming majority (median 0.0009 eV/atom) — strong agreement between the fine-tuned MACE potential and each source's own DFT. The worst outliers were almost all JARVIS candidates already flagged far from the convex hull by DFT — MACE's independent judgment corroborates the source thermodynamic data rather than contradicting it.

The classifier-confidence ranking and MACE relaxation-consistency signal were merged into one final ranking (converged → hull-stable → smallest MACE relaxation shift → classifier confidence). The top 30 are written to `outputs/phase4_shortlist.csv`, ready for Phase 4 simulation. The top candidate is **mp-1520073** (SrCaFeBiO₆, FM, ground-truth label, cubic, noncentrosymmetric, contains Bi, 0.089 eV/atom above hull, zero relaxation shift).

5. Phase 3 — MACE foundation-model fine-tuning

PHASE 3 · COMPLETE

MACE's `small` foundation model was fine-tuned on the harmonized dataset using multihead training: a `Default` head learns the new magnetic-materials data while a `pt_head` replays a 20,000-configuration sample of the original pretraining distribution, to fine-tune on the new domain without catastrophically forgetting the foundation model's general chemistry.

Setting	Value
Foundation model	MACE-small
Precision	float32 (switched from float64 default — see below)
Batch size	32
Max epochs	60 (SWA from epoch 45, patience 15)
Loss weighting	energy 1.0 / forces 100.0 / stress 1.0 (Stage 1); energy 1000 / forces 100 / stress 10 (Stage 2, post-SWA)
Replay set	20,000 foundation-distribution configurations (<code>pt_head</code>)

Performance fix

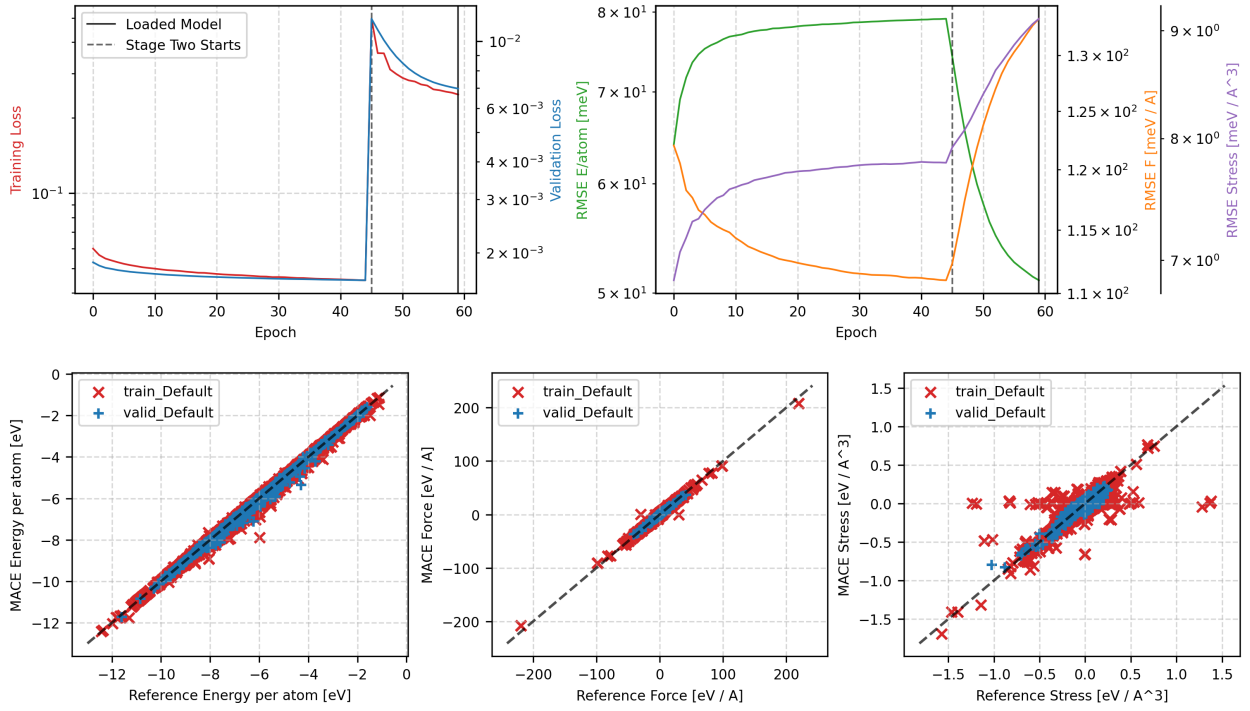
The initial run defaulted to float64 precision and single-threaded data loading (silent defaults, not deliberate choices), running at ~32 min/epoch (~32h projected). Switching to float32 with 8 parallel data-loading workers cut this to ~6.3 min/epoch — a 5× speedup, confirmed stable across all 60 epochs.

All 60/60 epochs finished successfully; both Stage 1 (epoch 44) and Stage 2/SWA (epoch 59) checkpoints were saved:

Checkpoint	Split	Energy RMSE	Forces RMSE	Stress RMSE
Stage 1 (epoch 44)	train	75.2 meV/atom	104.0 meV/Å	11.6
Stage 1 (epoch 44)	valid	79.1 meV/atom	111.0 meV/Å	7.8
Stage 2 / SWA (epoch 59)	train	41.7 meV/atom	122.6 meV/Å	12.8
Stage 2 / SWA (epoch 59)	valid	51.1 meV/atom	133.4 meV/Å	9.1

The Stage 2 loss reweighting markedly improved energy accuracy (valid RMSE_E 79.1 → 51.1 meV/atom) at a small cost to force/stress accuracy, as intended. Native training diagnostics below (loss curves, RMSE vs. epoch, and predicted-vs-reference parity plots for energy/forces/stress):

Model loaded from epoch 59 (Default head)



Default head, full run (epoch 0–59). Vertical line marks the Stage 2/SWA transition (epoch 45).

Partial-forgetting signal, resolved

During Stage 1, the `pt_head`'s replay-distribution energy RMSE drifted from 39.6 up to a peak of 134.4 meV/atom at epoch 43 ($\sim 3.4\times$ degradation) — genuine partial forgetting. The Stage 2 reweighting largely corrected this: `pt_head` RMSE recovered to 54.2 meV/atom by epoch 58, close to its starting value.

6. Phase 4a — Exploratory spin dynamics

PHASE 4a · COMPLETE (exploratory)

One candidate interfacial system — a Pd/Fe/Ir(111)-type stack, chosen for its interfacial DMI — was simulated with LAMMPS-SPIN (exchange + DMI + Zeeman/precession + Langevin thermostat + LBFGS minimization).

Solver bug found, root-caused, and fixed

A dynamics-integration segfault was isolated via systematic one-variable-at-a-time bisection (11 reproducer scripts) to a genuine upstream LAMMPS bug: `pair_style hybrid/overlay` segfaults whenever it wraps exactly two SPIN-package pair substyles. Confirmed against LAMMPS's own official `examples/SPIN/bfo` (which combines three substyles and runs cleanly on the same build). **Workaround:** pad the production Hamiltonian with a third, physically-inert substyle — verified against the real 600-atom system before adopting.

Findings

- Spin-spiral wavelength consistent across 5 independent seeds (5.0–5.8 nm, mean ~5.4 nm) — a genuine equilibrium property, not a quench artifact.
- That wavelength is ~20% longer than the ~4.5 nm target reported in reference literature for this system.
- Propagation direction is not consistently aligned to the crystallographically preferred (Γ –K) directions across seeds — indicative of a multi-domain outcome.
- Most likely cause: the DMI term is currently coded as a single uniform bond-independent vector, whereas this system physically requires a bond-direction-dependent (Fert–Levy) DMI form, which LAMMPS's stock `spin/dmi` pair style cannot express via one `pair_coeff`. Parked pending a priority call.

A refined 33-run finite-temperature scan (190–245 K, bracketing the reference paper's $T_c \approx 214$ K, run via single-core task-level parallelism after MPI domain decomposition was found structurally incompatible with this system's quasi-2D geometry) completed with no errors. Energy rises smoothly and monotonically with T as expected; the heat-capacity-like signal shows a mild decrease around 210–220 K rather than a sharp peak — a real but inconclusive signal, most likely because the 2 ps production window is too short to resolve critical slowing-down cleanly.

7. Phase 4b — Real DFT magnetic-ordering validation

PHASE 4b · IN PROGRESS · 24 OF 26 CANDIDATES COMPLETE

7.1 Why this check exists

Generalizing Phase 4a's spin dynamics to new shortlisted candidates requires per-candidate exchange (J) and DMI (D) constants that nothing in this pipeline can honestly produce — the fine-tuned MACE model was never trained with per-site `magmom` as an input feature, so it cannot distinguish FM from AFM orderings of the same atomic structure. Separately, the screening pipeline (§4) treats Materials Project's stored magnetic ordering as ground truth (`ordering_source=ground_truth`, `confidence=1.0`) wherever MP is the source database — an assumption this project had no way to check until real DFT compute became available on the account's HPC clusters (`paramsanganak` , IIT Kanpur).

7.2 Method

The same collinear magnetic-ordering energy-mapping approach behind Materials Project's own magnetic-ordering workflow (Horton, Montoya & Persson, *npj Comput. Mater.* 2019):

1. Take the candidate's relaxed structure from the harmonized dataset.
2. Enumerate symmetry-distinct FM/AFM spin configurations with `pymatgen.MagneticStructureEnumerator` (backed by `enumlib`).
3. Generate VASP inputs with `MPRelaxSet` — MP's own GGA+U, ENCUT (520 eV), and k-point convention (e.g. Fe gets `LDAUU=5.3` eV, matching MP exactly) — overriding only `MAGMOM` per ordering.
4. Run each configuration to convergence (48 cores/job). The original 6 candidates ran on `paramsanganak` (VASP 5.4.4); all later candidates ran on `paramrudra` (VASP 6.4.3) after `paramsanganak` became unavailable mid-project.
5. Compare converged total energies per atom; the lowest is the DFT ground state.

This keeps results directly comparable to the ground-truth MP energies each candidate was screened against, rather than being an arbitrary DFT recipe. 85 individual calculations (across the 24 complete candidates) converged cleanly — every `OUTCAR` reports "reached required accuracy". 2 candidates are still running: mp-1173143 (1 of 4 configs converged) and mp-682554 (1 of 4 configs converged, a sulfide with no LDA+U applied — unrelated to the tungsten issue below).

Infrastructure built for this check

POTCAR pseudopotentials are license-restricted and were never copied into this repository — only symbol names are generated locally; actual POTCAR files are concatenated remotely from the account's licensed pseudopotential library. Along the way, two environment bugs were found and fixed: a stale Intel compiler module name after a cluster module-tree reorganization (fixed by sourcing `compilervars.sh` / `mpivars.sh` directly), and a default MPI stack-size limit too small for VASP's non-local projector routines at 48-way parallelism (fixed with `ulimit -s unlimited`).

Hubbard U on tungsten removed for 5 candidates (methodological deviation)

All 5 W-bearing candidates (mp-754090, mp-755203, mp-769471, mp-775194, mp-850225) initially failed to converge under MP's standard GGA+U value (`U=6.2` eV on W): W here is formally W6+, nominally d0, and a

large U on a near-empty d-shell leaves the SCF density with no restoring force, so it oscillates indefinitely instead of converging. Confirmed with 11 independent sandbox tests; the only working fix is removing U from W entirely (`LDAUL=-1`), applied uniformly across all orderings of a compound. This converged all 18 production orderings cleanly on the first attempt (verified against false convergence via `rms(c)` , not just the "EDIFF is reached" banner). The within-compound FM-vs-AFM verdict for each of these 5 is unaffected, but their absolute energies are not on the same U footing as MP's database or the other 19 candidates here.

7.3 First case: mp-861559 (TaFeO₄)

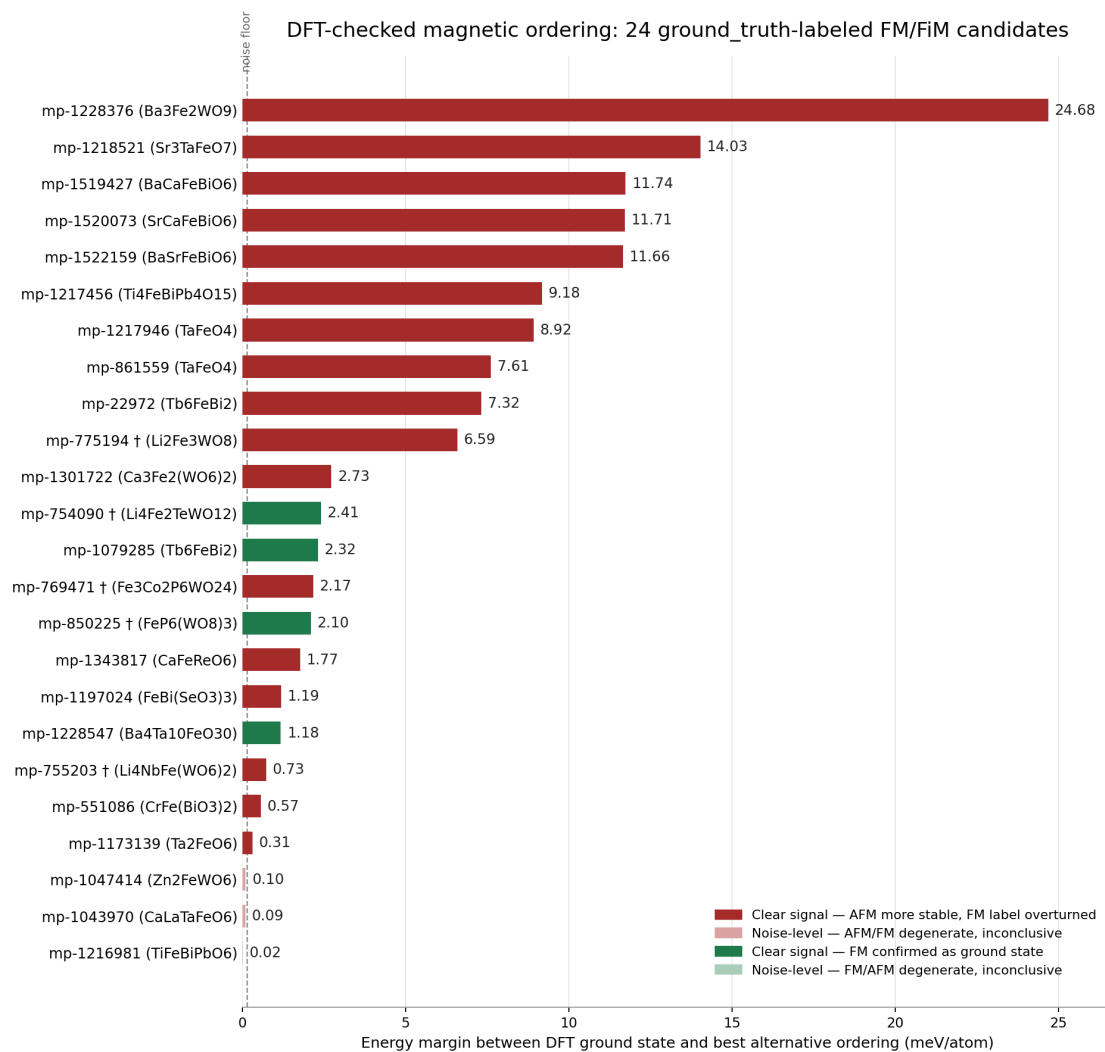
Chosen first because its 6 Fe sites in one 36-atom cell allow AFM ordering enumeration without a supercell. FM + 3 symmetry-distinct AFM orderings were run to convergence:

Ordering	TOTEN (eV)	mag (μ_B)	ΔE vs. FM
FM (MP's recorded label)	-304.60011	30.0	—
AFM-1	-304.87416	0.0	-0.274 eV
AFM-2	-304.78236	0.0	-0.182 eV
AFM-3	-304.74876	0.0	-0.149 eV

Sanity check on the setup itself: this run's converged FM magnetization (29.99998 μ_B) matches MP's own recorded value (30.0022251 μ_B) almost exactly — the local calculation reproduces MP's own result, so the AFM finding is not a convention mismatch. All three AFM orderings tested come out lower than FM; AFM-1 is the ground state, 0.274 eV/cell (≈ 45.7 meV/Fe) below FM. **The classifier's FM label for mp-861559 is not confirmed by DFT.** This one result argued for extending the same check to more shortlisted candidates.

7.4 Extending to the full 24-candidate batch

The remaining `ordering_source=ground_truth, confidence=1.0` entries from `outputs/phase4_shortlist.csv` were run through the identical pipeline. 24 candidates (85 calculations total) have completed; 2 more (mp-1173143, mp-682554) are still running. Five of the 24 required removing Hubbard U from tungsten to converge — see the callout above.



Candidate	Formula	Label	DFT ground state	Margin (meV/atom)	Signal
mp-1228376	Ba ₃ Fe ₂ WO ₉	FM	AFM	24.68	clear — overturned
mp-1218521	Sr ₃ TaFeO ₇	FM	AFM	14.03	clear — overturned
mp-1519427	BaCaFeBiO ₆	FM	AFM	11.74	clear — overturned
mp-1520073	SrCaFeBiO ₆	FM	AFM	11.71	clear — overturned
mp-1522159	BaSrFeBiO ₆	FM	AFM	11.66	clear — overturned
mp-1217456	Ti ₄ FeBiPb ₄ O ₁₅	FM	AFM	9.18	clear — overturned
mp-1217946	TaFeO ₄ (entry 2)	FM	AFM	8.92	clear — overturned
mp-861559	TaFeO ₄ (entry 1)	FM	AFM	7.61	clear — overturned
mp-22972	Tb ₆ FeBi ₂	FiM	AFM	7.32	clear vs FM, doesn't test FiM
mp-775194 †	Li ₂ Fe ₃ WO ₈	FM	AFM	6.59	clear — overturned

mp-1301722	Ca ₃ Fe ₂ (WO ₆) ₂	FM	AFM	2.73	clear — overturned
mp-754090 †	Li₄Fe₂TeWO₁₂	FM	FM	2.41	clear — confirmed
mp-1079285	Tb₆FeBi₂ (entry 2)	FM	FM	2.32	clear — confirmed
mp-769471 †	Fe ₃ Co ₂ P ₆ WO ₂₄	FM	AFM	2.17	clear — overturned
mp-850225 †	FeP₆(WO₈)₃	FM	FM	2.10	clear — confirmed
mp-1343817	CaFeReO ₆	FM	AFM	1.77	clear — overturned
mp-1197024	FeBi(SeO ₃) ₃	FM	AFM	1.19	clear — overturned
mp-1228547	Ba₄Ta₁₀FeO₃₀	FM	FM	1.18	clear — confirmed
mp-755203 †	Li ₄ NbFe(WO ₆) ₂	FM	AFM	0.73	clear — overturned
mp-551086	CrFe(BiO ₃) ₂	FiM	AFM	0.57	clear vs FM, doesn't test FiM
mp-1173139	Ta ₂ FeO ₆	FM	AFM	0.31	clear — overturned
mp-1047414	Zn ₂ FeWO ₆	FM	AFM	0.10	noise-level
mp-1043970	CaLaTaFeO ₆	FM	AFM	0.09	noise-level
mp-1216981	TiFeBiPbO ₆	FM	FM	0.02	noise-level

Breakdown of the 24

17 clearly overturned to AFM (0.31–24.68 meV/atom); **4 clearly confirmed** as FM (mp-1079285, mp-1228547, mp-754090 †, mp-850225 †; 1.18–2.41 meV/atom); **3 noise-level ties** (≤ 0.10 meV/atom: mp-1047414, mp-1043970, mp-1216981) sitting well inside the EDIFF=1e-6 eV electronic-convergence tolerance (MPRelaxSet default) — read as "FM and AFM degenerate to DFT precision," not as confirming either ordering. Among the 21 that resolve clearly, "AFM overturns FM" outnumbered "FM confirmed" 17 to 4.

7.5 Conclusion

4 of 24 `ground_truth`, `confidence=1.0` FM labels checked against real DFT are confirmed as the true magnetic ground state. 17 are clearly overturned to AFM; 3 are statistically tied and uninformative (2 candidates are still running). This is no longer a single anecdote — it is a consistent pattern showing the same MP labeling gap recurring across unrelated compositions and space groups: MP's routine per-entry pipeline relaxes from a single initial magnetic-moment guess and never enumerates AFM alternatives, so its stored "ground truth" reflects whatever configuration that guess happened to converge to.

Caveat on sample composition

This is 24 candidates, not a random sample — all were pulled from the top of the DMI-filtered shortlist (noncentrosymmetric, heavy-SOC-element, net-moment structures). It is plausible this structural family is more AFM-prone than the `ground_truth` population as a whole, which would mean the true project-wide mislabeling rate is lower than "17/24." What this sample *does* establish firmly: the label cannot be taken at face value for candidates in this specific shortlisted population without an actual DFT check — and this population is exactly what Phase 2 (§4) and Phase 4a (§6) draw from.

8. Summary of open items

Item	Status
MAGNDATA rescrape (2,414 entries)	Complete — 1,919 collected, 495 permanent upstream failures
Structure-aware classifier (v2)	Complete — 56.7% → 67.0% accuracy
OQMD / 2DMatPedia / NOMAD acquisition	Complete — 3,178 records added; OQMD bug fixed
Candidate screening pipeline	Complete — 1,647 pass funnel, 298 relaxed, top 30 shortlisted
MACE fine-tuning (60 epochs)	Complete — Stage 1 + Stage 2/SWA checkpoints saved
Exploratory spin dynamics (Pd/Fe/Ir)	Complete — LAMMPS bug fixed, wavelength/direction findings in hand
Real DFT magnetic-ordering validation	24/26 shortlisted candidates complete (24/1,647 overall) — 4 of 24 confirmed, 17 overturned, 3 tied
Phase 4 candidate selection (how many of 30 to simulate)	Shortlist ready; with 17/24 FM labels overturned, DFT findings still argue for caution before committing spin-dynamics compute to unvalidated candidates
C2DB acquisition	Not started
Bond-dependent (Fert–Levy) DMI implementation	Parked, needs a priority call

9. Recommendation for discussion

1. **Headline result:** real DFT check of 24 shortlisted `ground_truth` FM labels finds 17 clearly wrong (AFM ground state), 3 inconclusive/degenerate, and 4 confirmed as FM.
2. **Implication:** the classifier's/MP's ordering label should be treated as a *screening prior*, not ground truth, for this shortlist's structural family (noncentrosymmetric, heavy-SOC, Fe-magnetic-site compounds). Worth deciding whether to re-flag affected shortlist entries, or whether this changes candidate ranking — the FM/AFM label doesn't currently feed the DMI funnel directly, but does feed `ordering_final`, which is used to select "net-moment" candidates in `screen_candidates.py`, and AFM is excluded from that funnel entirely.
3. **Open question:** is 24 of 1,647 candidates — still all drawn from the top of the DMI-filtered shortlist — enough evidence to act on, or does this need a larger, more representative validation batch, e.g. a random sample stratified across the whole `ground_truth` population, before changing how the pipeline treats these labels?
4. **Separately:** the Fert–Levy bond-dependent DMI implementation (§6) is parked pending priority, and the spin-dynamics T_c signal (§6) is inconclusive at the current 2 ps run length — both are candidates for the next block of compute time once the DFT-validation question above is settled.

Raw data references

- Classifier: `outputs/lightgbm_v2_eval_summary.json`, `outputs/lightgbm_v2_ablation.json`, `outputs/lightgbm_v2_feature_importance.csv`
- Screening: `outputs/candidate_screened_full.csv`, `outputs/candidate_relaxation_ranked.csv`, `outputs/phase4_shortlist.csv`
- DFT validation: `outputs/dft_ordering_validation.csv` (85 rows, 24 complete candidates, 2 more running); staged calculations under `data/dft_staging/<candidate_id>/{0_fm,1_afm,...}/results/`
- Detailed single-candidate and full 24-candidate writeups: `docs/dft_validation_mp-861559.md`, `docs/dft_validation_ground_truth_labels.md`